

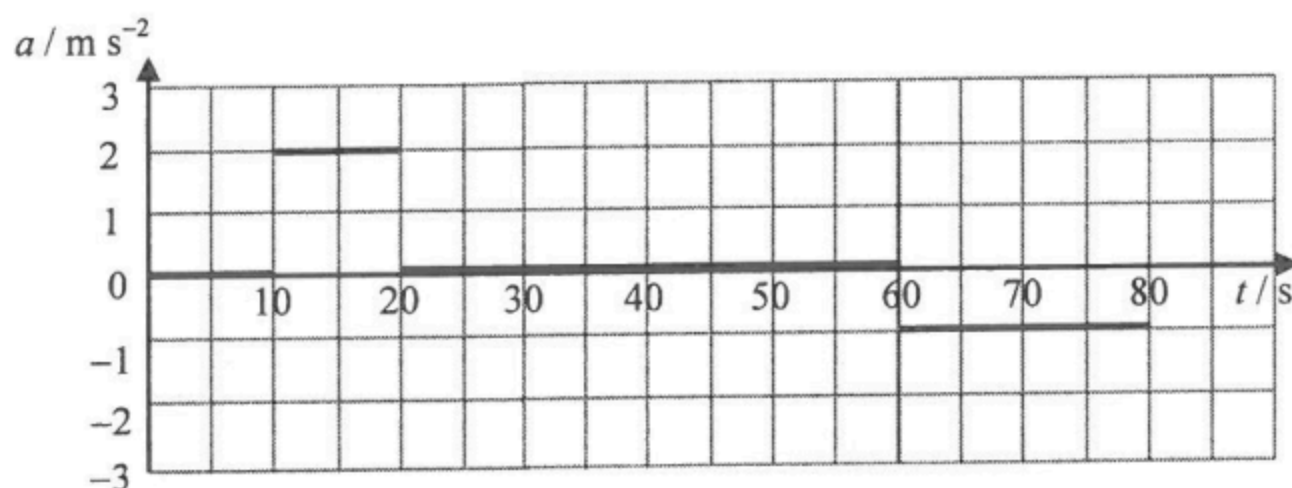
3. (a) 0 – 10 s: constant / uniform acceleration
10 – 80 s: constant velocity / uniform motion

1A
1A
2
1A
1A
2

- (b) (i) Car B.
It is represented by the steepest slope between 10 s and 20 s for B.

$$a = \frac{20 - 0}{20 - 10} = 2 \text{ m s}^{-2}$$

(ii)



Uniform acceleration between 10 s and 20 s & uniform deceleration between 60 s and 80 s.

Or Either acceleration or deceleration correct.

All correct.

1A
1A
1A
2

- (c) (i) Total area of A by 20 s = $\frac{(10 + 20)}{2} \times 15 = 225 \text{ m}$
Total area of B by 20 s = $\frac{(10 \times 20)}{2} = 100 \text{ m}$
Separation of A & B, $s_{AB} = 225 \text{ m} - 100 \text{ m} = 125 \text{ m}$

1M
1A
2

- (ii) Let B meet A at time $(20 + T) \text{ s}$,
 $(v_B - v_A) \times T = s_{AB}$
 $(20 \text{ m s}^{-1} - 15 \text{ m s}^{-1}) \times T = 125 \text{ m}$
 $T = 25 \text{ s}$
 $\therefore B \text{ catches up with A at } t = (20 + 25) \text{ s} = 45 \text{ s}.$

Or $s_A + v_A \times T = s_B + v_B \times T$
 $225 \text{ m} + 15 \text{ m s}^{-1} \times T = 100 \text{ m} + 20 \text{ m s}^{-1} \times T$
Or Accept using graphical method and read the answer from the graph

1M
1A
1M
1A
2

- (d) Let F_A and F_B be the driving forces developed by the engines of A and B respectively.

Ratio of power developed by cars A and B,

$$\begin{aligned} P_A : P_B &= F_A \times v_A : F_B \times v_B \\ &= v_A^2 \times v_A : v_B^2 \times v_B \\ &= 3^3 : 4^3 = 27 : 64 \end{aligned}$$

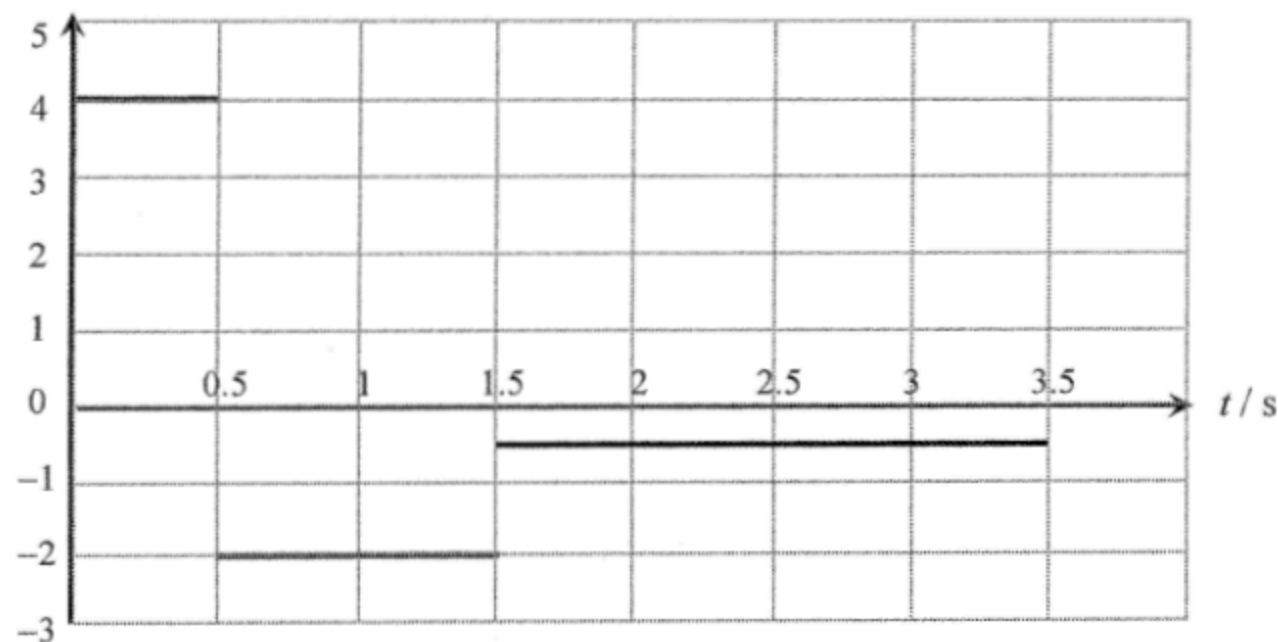
1M
1A
2

4. (a) The block decelerates uniformly (until at rest or its velocity is zero at $t = 1.5$ s).
It then moves with uniform acceleration down the plane (until $t = 3.5$ s).

(b) (i)
$$a_2 = \frac{-1 - 0}{3.5 - 1.5}$$

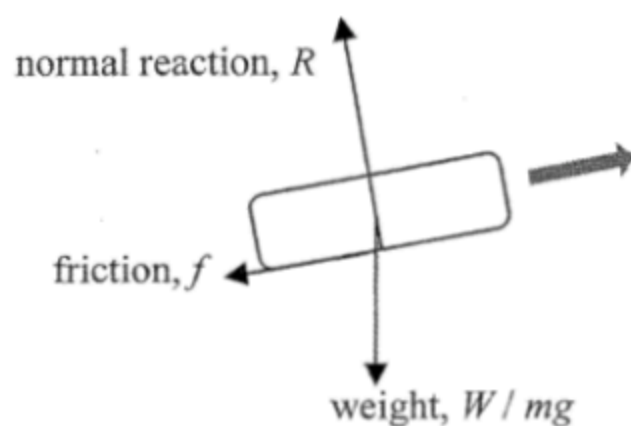
$$= 0.5 \text{ m s}^{-2}$$

(ii) $a / \text{m s}^{-2}$



Either acceleration correct
All correct

(c)



Friction correctly indicated
All correct

(d) Moving upward : $-mg \sin \theta - f = ma$
 $-(1)(9.81) \sin \theta - f = (1)(-2) \dots \dots \dots \textcircled{1}$

Moving downward : $-mg \sin \theta + f = ma'$
 $-(1)(9.81) \sin \theta + f = (1)(-0.5) \dots \dots \dots \textcircled{2}$

$\textcircled{2} - \textcircled{1} : 2f = 1.5$
 $f = 0.75 \text{ N}$

(Note: $\theta = 7.32^\circ$)

1A
1A
2
1M
1A
2

1A
1A
2
1A
1A
2
1M
1M
1A
3

3. (a) $a = \frac{6-0}{2-0}$
 $= 3 \text{ m s}^{-2}$ (downwards)

1M

1A

2

(b) A: 395 N B: 569 N C: 685 N

1A

In stage B, balance reading = weight (Newton's 1st law),

$$mg = m \times 9.81 \text{ m s}^{-2} = 569 \text{ N}$$

$$m = 58.0 \text{ kg}$$

1M

1A

For $g = 10 \text{ m s}^{-2}$, $m = 56.9 \text{ kg}$

Or In stage A, according to Newton's 2nd law
 $(569 - 395) \text{ N} = ma = m(3 \text{ m s}^{-2})$
 $m = 58.0 \text{ kg}$

1M

1A

3

(c) (i) For stage C, by Newton's 2nd law,

$$F = ma$$

$$(569 - 685) \text{ N} = (58.0 \text{ kg}) a$$

$$a = -2 \text{ m s}^{-2}$$

1M

For $g = 10 \text{ m s}^{-2}$, $m = 56.9 \text{ kg}$,
 $a = -2.04 \text{ m s}^{-2}$

$$\text{Thus, } a = \frac{0-6}{T-12} = -2 \text{ m s}^{-2}$$

$$T = 15 \text{ s}$$

1M

2

(ii) Height $\approx$ displacement of lift = area under graph

$$= \frac{(12-2)+15}{2} \times 6$$

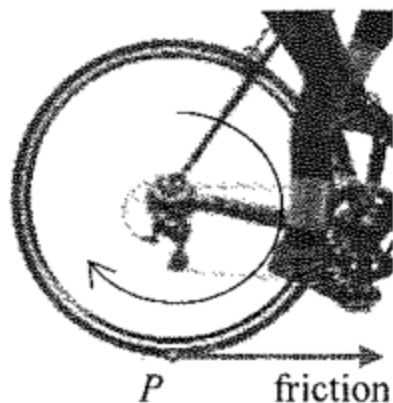
$$= 75 \text{ m}$$

1M

1A

2

3. (a)



1A

1

- (b) Mechanical power delivered (for work done against the resistive force)

$$\begin{aligned} P &= Fv \\ &= 17.0 \times 8.0 \\ &= 136 \text{ W} \end{aligned}$$

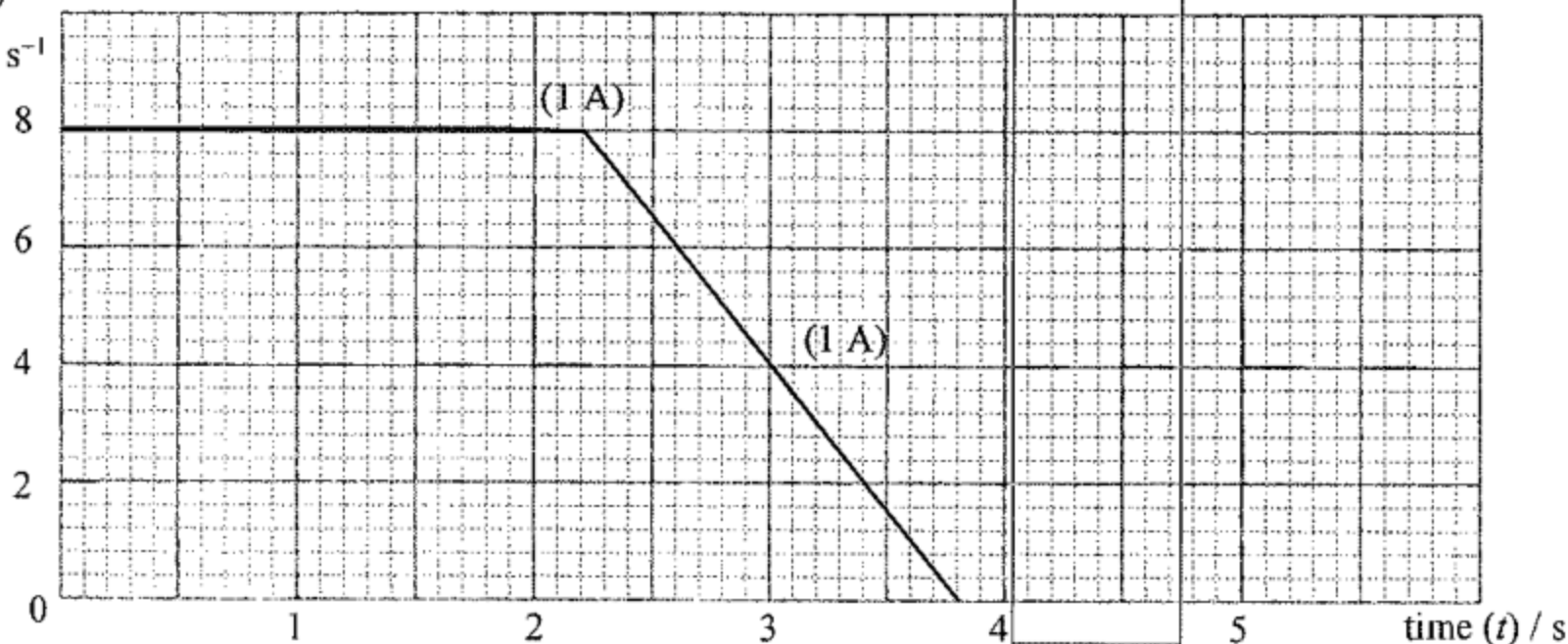
1M

1A

2

- (c) (i)

velocity / m s⁻¹



2

- (ii) Deceleration: $\left| \frac{0-8}{3.8-2.2} \right| = 5.0 \text{ m s}^{-2}$
Resistive force = 65×5.0
 $= 325 \text{ N}$

1M

e.c.f. from (c)(i)

1A

Alternative method:

$$\text{Braking distance} = \frac{1}{2}(8)(3.8 - 2.2) = 6.4 \text{ m}$$

$$Fs = \frac{1}{2}mv^2 - 0$$

$$F(6.4) = \frac{1}{2}(65)(8)^2$$

$$F = 325 \text{ N}$$

1M

1A

2

- (iii) Total distance travelled after $t = 2.0 \text{ s}$
 $=$ area under the graph from $t = 2.0 \text{ s}$ to $t = 3.8 \text{ s}$
 $= \frac{0.2+1.8}{2} \times 8.0 = 8.0 \text{ m}$

1M/1A

$$\begin{aligned} \text{Distance between the bicycle and the obstacle when the} \\ \text{bicycle stops} &= 9 - 8 \\ &= 1 \text{ m} \end{aligned}$$

1M

1A

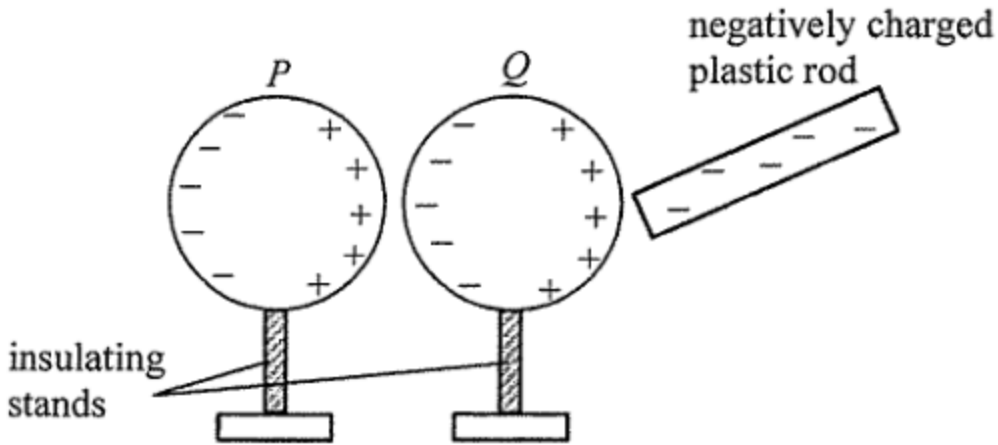
3

- (d) The padded cushion can increase the contact time of the knee and the ground such that a smaller force acts on the knee.

1A

1A

2

9. (a)		2A	2	1A+1A	2	1A	1M	1M for the correct procedure to identify the polarity of the charged rod.
(b) No, it remains unchanged as charging is through induction (without any contact or transfer of charges).								
(c) Hold / bring the unknown charged rod (assume positive) near sphere P, which carries positive charges, without touching it.								
If P is repelled by the rod, the rod should carry positive charges. <u>And / Or</u> If P is attracted by the rod, the rod should carry negative charges.								

3. (a) The object is moving to the left with constant deceleration and (momentarily) stops at $t = 4$ s.

(b) Displacement = total area of the graph

$$= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10 - 4)$$

$$= 25 \text{ m}$$

Position of the object is 25 m from point O on the right side.

(c) Net force acting on the object

$$= ma$$

$$= 0.4 \times \frac{15 - (-10)}{10 - 0}$$

$$= 1 \text{ N (towards right)}$$

1A

2

1M

1A

2

1M

1A

2

Or

$$v^2 = u^2 + 2as$$

$$15^2 = (-10)^2 + 2 \times \frac{15 - (-10)}{10 - 0} \times s$$

$$s = 25 \text{ m}$$